

CS2601 Linear and Convex Optimization

Homework 1

Due: 2024.10.17

1. Determine if the given vectors $\mathbf{v}$ and $\mathbf{u}$ are eigenvectors of $\mathbf{A}$? If yes, find the eigenvalue of $\mathbf{A}$ associated to the eigenvector.

$$\mathbf{A} = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

2. Find the gradients and the Hessians of the following functions,

(a). $f(\mathbf{x}) = \frac{1}{2}\|\mathbf{x}\|^2$

(b). $f(\mathbf{w}) = \frac{1}{2}\|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2$, where $\mathbf{X}, \mathbf{y}, \lambda$ are known constants.

3. For what value of α is the following matrix $\mathbf{A}$ positive semidefinite?

$$\mathbf{A} = \begin{pmatrix} 3 & -1 & 2 \\ -1 & 1 & \alpha \\ 2 & \alpha & 2 \end{pmatrix}$$

4. Find the stationary points of the following functions, and determine if they are local minima, local maxima, or neither.

(a). $f(x_1, x_2, x_3) = 2x_1^2 + \frac{5}{2}x_2^2 + 3x_3^2 + 2x_1x_2 - 2x_2x_3 + 2x_1 - 3x_2 + 2x_3$

(b). $f(x_1, x_2, x_3) = \frac{1}{2}x_1^2 + x_2^2 - \frac{3}{2}x_3^2 + 2x_1x_2 - x_2x_3 + x_2 - 3x_3$

5. **Logistic regression.** Recall the objective function of logistic regression is the following negative log likelihood,

$$f(\mathbf{w}) = \sum_{i=1}^m \log(1 + e^{-y_i \mathbf{x}_i^T \mathbf{w}}),$$

where $(\mathbf{x}_i, y_i) \in \mathbb{R}^n \times \{-1, +1\}$ is the i -th data point. We have absorbed the bias term b into $\mathbf{w}$ by appending an extra 1 to each $\mathbf{x}_i$.

- (a). Suppose the dataset $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)\}$ is strictly linearly separable in the sense that there exists a $\mathbf{w}_0$ such that

$$y_i \mathbf{x}_i^T \mathbf{w}_0 > 0, \quad \forall i = 1, 2, \dots, m.$$

Does f have a global minimum in this case? Explain your answer.

- (b). Suppose the dataset is not linearly separable in the sense that for any $\mathbf{w}$, there exists an $i_0 = 1, 2, \dots, m$ such that

$$y_{i_0} \mathbf{x}_{i_0}^T \mathbf{w} < 0.$$

Show that f has a global minimum by completing the following steps.

- i) Show

$$f(\mathbf{w}) \geq h(\mathbf{w})$$

where

$$h(\mathbf{w}) = \max_{1 \leq i \leq m} -y_i \mathbf{x}_i^T \mathbf{w}.$$

- ii) Let $S = \{\mathbf{w} : \|\mathbf{w}\| = 1\}$ be the unit sphere. Show that $h(\mathbf{w})$ has a global minimum $\mathbf{w}_0$ on S and $C \triangleq h(\mathbf{w}_0) > 0$. You can assume the fact that h is continuous, which can be proved by induction and the identity $\max\{a, b\} = \frac{a+b+|a-b|}{2}$.

- iii) Show

$$h(\mathbf{w}) \geq C \|\mathbf{w}\|, \quad \forall \mathbf{w}$$

- iv) Show f has a global minimum.

- (c). Find $\nabla f(\mathbf{w})$.